FF-S-85C
AMENDMENT-1
February 27, 1973

FEDERAL SPECIFICATION

SCREW, CAP, SLOTTED AND HEXAGON HEAD

This amendment, which forms a part of Federal Specification FF-S-85C, dated May 5, 1969, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

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Paragraph 2.1: Add the following Federal Specification: "PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of."

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Paragraph 2.1: Delete "MIL-H-3982" and title.

Paragraph 2.1: Add the following Military Specification: "MIL-F-18240 - Fastener, Externally Threaded, 250°F. Self-Locking Element For."

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Paragraph 2.2: Delete "United States of America Standards Institute Standards:" and substitute "American National Standards Institute (ANSI) Standards."

Paragraph 2.2: Delete "USAS" and substitute "ANSI" (two places).

Paragraph 2.2: Delete "United States of America Standards Institute, 10 East 40th Street, New York, New York 10016" and substitute "American National Standards Institute, 1430 Broadway, New York, NY 10018".

Paragraph 2.2: Add the following ASTM Publication: "B154 - Mercurous Nitrate Test for Copper and Copper Alloys"

FSC 5305

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Add the following new paragraphs:

3.1.5.1 Stress relief. Cold worked naval brass screws shall be stress relieved by heating the screws to a temperature of 600°F. and holding at that temperature for a minimum of 1 hour.

3.1.6 Self-locking element. Non-metallic, self-locking elements shall be in accordance with MIL-F-18240.

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Table IA - For Naval Brass, under Composition, add "alloy 462" and delete "Composition 1": Under Condition, add "hard" opposite alloy 462.

Table IA - For Nickel-Copper Alloy, under Condition, delete "hot finished": Under Elongation, delete "30" and substitute "20".

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Paragraph 3.3: Delete "USAS" and substitute "ANSI".

Paragraph 3.4.3: Delete in its entirety and substitute the following:

3.4.3 Cadmium plating. When specified (see 6.2), cap screws shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2. (Cadmium plating in accordance with QQ-P-416, Type II, Class 3 may be furnished until 30 March 1975).

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Paragraph 3.4.7: Delete in its entirety and substitute the following:

3.4.7 Black chemical. When specified (see 6.2), brass cap screws shall be treated with a black chemical finish in accordance with MIL-F-495.

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Paragraph 3.8: Delete the first two sentences and substitute the following: "Steel cap screws shall be marked with the grade identification symbol conforming to SAE J429 and applicable documents. In addition, all cap screws shall be marked with the manufacturer's identification symbol."

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Add the following to Table III - Classification of Defects:

"105 Self-locking element missing, when applicable (3.1.6) Visual"

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Paragraph 4.2.5: Delete "MIL-H-3982" and substitute "PPP-H-1581".

Add the following new paragraph:

4.2.6 Sampling for mercurous nitrate test. When specified (see 6.2), a random sample shall be taken from each lot of cold-worked naval brass screws in accordance with MIL-STD-105, Inspection Level S-1. Any one defect shall cause rejection of the represented lot.

Paragraph 4.3.2: Delete "MIL-H-3982" and substitute "PPP-H-1581".

Paragraph 4.4.1: Delete "4.4.9" and substitute "4.4.11".

Paragraph 4.4.3: Add the following sentence: "Unless otherwise specified (see 6.2), Grade 2, low carbon steel screws shall not be subjected to this test."

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Add the following new paragraphs:

4.4.10 Self-locking. Self-locking screws having non-metallic elements shall be tested in accordance with MIL-F-18240.

4.4.11 Mercurous nitrate (stress relief) test. The contractor shall furnish the Government certification that cold worked naval brass screws have been subjected to the stress relief treatment specified in 3.1.5.1. When specified (see 6.2), samples taken as specified in 4.2.6 shall be subjected to a mercurous nitrate test conducted in accordance with ASTM B154. There shall be no evidence of cracks.

Paragraph 5.1.1: Delete "MIL-H-3982" and substitute "PPP-H-1581" (two places).

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Paragraph 6.2 - Add the following:

"(p) Mercurous nitrate test, if required (4.4.11)."

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MILITARY CUSTODIANS:

Army - WC
Navy - SH
Air Force - 82

Reviewer Activites:

Army - AV, MU
Navy - None
Air Force - None
DSA - IS
NS

User Activities:

Army - AT, GL, ME, MI
Navy - AS, MC, OS, YD
Air Force - None

Preparing activity:

Army - WC

Project No. 5305-1147

CIVIL AGENCY COORDINATING ACTIVITIES:

COMMERCE - NBS
GSA-FSS
NASA-MSF

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